Supplementary Material: Does Displayed Reasoning Drive the Answer? A Diagnostic Benchmark for Vietnamese Educational Reasoning

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Abstract

This supplementary document provides methodological protocols, human rating agreement metrics, detectability distributions, technical sensitivity audits, and knowledge probe evaluation outcomes accompanying the main paper.

A Rating Protocol Summary

Human review comprises three distinct protocols, each serving a separate methodological role: (i) *Pre-intervention chain rating*: Evaluates the question, answer options, and prefix through step 3 (without step 4 or model continuations) to assess whether corrupted and salted clean catch stimuli exhibit detectable reasoning defects (0–5 detectability) and whether the error necessitates a shifted final answer (binary answer relevance). Zero denotes no detected error, including on corrupted stimuli. (ii) *Placebo equivalence review*: Independently compares the original clean step 3 against its rewritten variant to verify meaning preservation (yes/no), ensuring the paraphrase does not alter logical assertions, factual scope, or intermediate conclusions. (iii) *Post-intervention disclosure evaluation*: Conducted separately on generated continuations to score explicit error acknowledgment and repair (0–3), providing ground-truth human calibration for automated disclosure judging without influencing pre-intervention chain validity.

B Concrete Matched Triplet Stimulus Example

Table 1 illustrates a canonical item (`S1-HS-01-MATH`, High School, Sequential Dependence S1) showing the original prompt, balanced options, and the three controlled step-3 prefix conditions alongside their expected continuation answers.

C Run-Level Data Schema

Each intervention generation stores its draw/item/chain identifiers, item version, arm, draw index, seed, requested and served model/provider, decoding settings, supplied prefix, generated continuation, visible response, parsed option, gold and propagation options, usage metadata, hashes, and protocol diagnostics. Triplet, problem type, level, and schema are joined from item metadata; they need not be duplicated in every draw row. Language compliance is checked with a rule-based classifier (Vietnamese diacritics and function words vs. English words) across all 9,450 non-empty continuations: 9,446 are Vietnamese, 4 are Vietnamese with a few isolated English words (e.g., “option, “substitution), and none is English. The stored fields permit reconstruction of cell-level answer probabilities and checks for duplicate draws.

D Rating Agreement

Agreement is calculated from the two independent human raters' scores across all 162 executed items in ratings/rated/. Missing/not-applicable relevance is excluded from binary agreement, not converted to a negative judgment. Clean catch trials form a separate pool, excluded from corruption-severity agreement; their false-alarm rate counts positive detectability judgments.

Human agreement uses paired corrupted-prefix and placebo ratings across all 162 executed items, evaluated independently by two raters under blinded evaluation protocols. Raters saw only the rating sheets, never model outputs; the ratings did not determine inclusion in the 162-item primary analysis or change any outcome label.

Quadratic-weighted detectability κ_w is 0.282 (62 exact matches and 35 differences of at least two points). For binary answer relevance across 160 valid pairs (2 pairs contain missing/not-applicable relevance where detectability was rated zero), raw observed agreement is 96.88% (155/160: 153 two-positive, 2 two-negative, and 5 conflicting pairs). Due to the high-prevalence paradox [1], where 97.2% of marginal ratings are positive, Cohen's κ is suppressed to 0.429; prevalence-adjusted measures confirm high concordance (PABAK = 0.938, Gwet's AC_1 = 0.967).

Placebo reviews have 162 paired human ratings ($\kappa = 0.568$, observed agreement 87.04%): 122 two-positive, 19 two-negative, and 21 conflicting pairs. A separate pool of 29 clean catch stimuli has 58 judgments and 3 false alarms (5.2%). Catches are excluded from corruption agreement.

Two positive detectability/relevance judgments plus two positive placebo reviews select 117 items (25 complete triplets); adding protocol completeness and probe success leaves 88 items (13 complete triplets). These review-based sets are sensitivity subsets (Appendix F), not the principal diagnostic sample or adjudicated scores.

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Component	Content (S1-HS-01-MATH, High School, Sequential Dependence S1)
Question	A customer purchases imported goods via an e-commerce platform with a customs-declared value of 800,000 VND. The import duty is 20% of the declared value. A fixed, untaxed customs handling fee is 300,000 VND. What is the total amount the customer must pay? *Vietnamese stem: "Một khách hàng đặt mua lô hàng qua sàn TMĐT có giá trị khai báo 800.000 đồng..."*
Options	**A [Gold]**: 1,260,000 VND **B**: 1,100,000 VND **C [Propagation]**: 1,320,000 VND **D**: 960,000 VND
Step 1 (Retrieval)	Declared value: 800,000 VND; duty rate: 20%; fixed customs handling fee: 300,000 VND.
Step 2 (Inference)	Total payment = Goods declared value + Import duty + Fixed handling fee.
Step 3: Clean	Import duty = $800,000 \times 20\% = 160,000$ VND. Component costs are 800,000 VND goods, 160,000 VND duty, and 300,000 VND handling fee. $\Longrightarrow$ **Continues to A (Gold: 1,260,000 VND)**
Step 3: Placebo	Applying the 20% duty rate to the declared value gives $800,000 \times 20\% = 160,000$ VND; the other components are unchanged. $\Longrightarrow$ **Continues to A (Gold: 1,260,000 VND)**
Step 3: Corrupted	Import duty is calculated as 20% on the combined sum of goods and handling fee: $(800,000 + 300,000) \times 20\% = 220,000$ VND. $\Longrightarrow$ **Continues to C (Propagation: 1,320,000 VND)**

Table 1: Canonical stimulus demonstration on S1-HS-01-MATH. Steps 1–2 establish the shared deductive framework; Step 3 manipulates logic while Step 4 is generated under unconstrained continuation. The matched theoretical science item (S1-HS-01-CHEM: identifying the excess reactant from molar ratios) and history item (S1-HS-01-HIS: determining which political event causally triggered a subsequent one) share the S1 sequential dependence schema with identical step-type sequence and manipulation position.

Problem type	n	κ_{severity}	$\kappa_{\text{relevance}}$
Computational	54	0.285	0.486
Theoretical	54	0.293	—
History/literature	54	0.265	0.381

Table 2: Agreement on corrupted-chain ratings across all 162 items. A dash indicates undefined κ when both reviews are constant. Raw detectability distributions appear in Appendix E.

E Detectability Distributions

Problem type	0	1	2	3	4	5
Computational	0	2	15	23	32	36
Theoretical	0	3	9	17	50	29
History/literature	2	0	13	36	35	22

Table 3: Counts of two raw human judgments per corrupted chain across all 162 executed items ($n = 54$ per problem type). They are not independent experimental units or adjudicated scores.

Mean detectability scores exhibit common support across problem types (3.79 ± 1.11 in computational, 3.86 ± 1.00 in theoretical, and 3.56 ± 1.06 in history/literature problems on the 0–5 scale; overall mean 3.73 ± 1.06). Correlation between item-level mean detectability and Δ_{CDE} is negligible ($r = -0.096$ overall), providing no evidence that propagation is confounded by perceived error salience (attenuated by $\kappa_w = 0.28$).

F Additional Sensitivity Analyses

All sensitivity subsets below use the same whole-triplet bootstrap (10,000 replicates; seed 20261013) as the principal estimate ($\Delta_{\text{CDE}} = 10.9$ points, [7.3, 14.8]; 162 items).

Human-screened. Requiring both raters to judge the corruption detectable and answer-relevant and both to confirm placebo meaning preservation selects 117 items (51 triplets, 25 complete) with $\Delta_{\text{CDE}} = 12.1$ points (95% interval: [8.2, 16.6]). This screen already excludes the one revised stimulus with unresolved propagation uniqueness.

Strict intersection. Further restricting to protocol-complete items whose knowledge probes all pass leaves 88 items (49 triplets) with $\Delta_{\text{CDE}} = 13.4$ points ([8.3, 19.0]); restricting to the 13 triplets complete under all criteria (39 items) gives $\Delta_{\text{CDE}} = 10.9$ points ([5.6, 16.5]).

Protocol and stimulus exclusions. The 140 protocol-complete items give $\Delta_{\text{CDE}} = 10.4$ points ([6.5, 14.9]). Excluding only the unresolved stimulus from the broad corpus leaves 161 items with $\Delta_{\text{CDE}} = 10.9$ points ([7.4, 14.8]); this exclusion concerns stimulus validity, not individual draws.

All subsets reproduce a positive corrupted-versus-placebo effect with intervals excluding zero. Raters never saw model outputs, so these screens use no outcome information.

G Exploratory Disclosure Ratings

The separate disclosure subset has 90 paired human scores: 76 corrupted and 14 clean. Quadratic-weighted κ is 0.530, with 68 exact matches. Problem-type values are 0.548 (28 continuations), 0.489 (30 continuations), 0.544 (32 continuations), respectively.

For corrupted continuations, the two raw reviews assign scores 0/1/2/3 to 54/19/2/1 and 59/0/0/17 cases; 2 receive scores ≥ 2 in both. Clean controls receive zero in both reviews. Separate judgments are retained, not averaged into consensus.

Variable	Coef (β)	Cluster SE	t-stat	p-value
Constant (History intercept)	+6.63	8.53	+0.78	0.440
q_i (History slope)	+8.32	10.08	+0.82	0.413
Computation dummy	+22.68	14.60	+1.55	0.126
Theoretical dummy	+26.70	19.55	+1.37	0.178
$q_i \times$ Computation	−37.70	17.24	−2.19	0.033
$q_i \times$ Theoretical	−38.49	24.31	−1.58	0.119

Table 4: Cluster-robust regression of Δ_{CDE} on q_i and domain interaction terms ($G = 54$ triplet clusters). The computation interaction has uncorrected $p = 0.033$ but is not significant after Bonferroni correction ($p = 0.066$) or by the omnibus test ($\chi^2(2) = 5.19$, $p = 0.075$).

We validate an automated LLM judge (`gpt-5.6-terra`) on the same 90-item disclosure subset. On the 14 clean controls, the automated judge achieves a 0% false-alarm rate (14/14 scored 0), matching both human raters. On the 76 corrupted continuations, the automated judge assigns scores 0/2/3 to 23/7/46 cases. Compared against Rater 2, the automated judge achieves 51.3% exact agreement, quadratic-weighted $\kappa = 0.246$, and Gwet's $AC_1 = 0.407$, capturing 17/17 human-identified explicit disclosures while adopting a more sensitive threshold that flags 36 continuations human raters scored 0. Compared against Rater 1, raw agreement is 28.9% ($\kappa_{quad} = 0.102$; $AC_1 = 0.101$). Corpus-wide automated disclosure scoring remains exploratory and does not alter the primary causal estimands.

Lexical Audit Validation

The 90-item Job B disclosure calibration subset (28 computational, 30 theoretical, 32 history/literature by problem type) contains 76 corrupted continuations and 14 clean controls. On these continuations, the corpus keyword classifier (regex list) flags 17/76 draws. Against Rater 1 (positive, $n = 22/76$) it has precision = 0.65, recall = 0.50; against Rater 2 (positive, $n = 17/76$), precision = 0.47, recall = 0.47. The broader package classifier flags 40/76 (precision/recall 0.40/0.73 and 0.33/0.76) and flags 60.9% of the corpus. Adjusting the flag rate by precision/recall gives explicit repair ≈ 26%–34% under either classifier, so the corpus rate (25.9% explicit repair, 74.1% silent override) is roughly unbiased, with silent override ≈ 66%–74%. Because verification keywords concentrate in arithmetic tasks, apparent rises in explicit repair on difficult quantitative problems may reflect recalculation density rather than true behavioral correction.

H Multivariable Interaction and Difficulty Moderation

Table 4 presents the cluster-robust multivariable regression model ($G = 54$ triplets) testing whether direct answerability (q_i) moderates corruption dependence (Δ_{CDE}) differently across problem types.

I Execution Assembly and Audit

The unified diagnostic assembly covers 162 items in 54 structurally complete triplets, yielding 9,720 records with all 20 scheduled draws per arm across clean, placebo, and corrupted conditions. All items adhere to the identical experimental protocol: three-arm continuation sampling ($k = 20$ draws each under deterministic seeds), automated constraint checks, and paired independent human review (corruption severity/relevance and placebo equivalence). No individual draws are selected on outcome.

Every unparseable answer remains in its 20-draw denominator as non-propagation (13 records across 9,720 draws). The broad diagnostic estimate retains records with minor protocol flags (the native-reasoning check flags 270 draws lacking reasoning tokens from premature <think> closure: 66 clean, 123 placebo, 81 corrupted), while a protocol-complete sensitivity removes entire items failing the 20-valid-draws-per-arm requirement. Bootstrap samples select whole triplets, preserving all arms and members. Machine-readable aggregate statistics, per-item counts, and anonymized checksums accompany the analysis. Paired independent human agreement covers all 162 executed items across both corruption-severity and placebo-equivalence evaluations.

Schema	Items	Triplets	Δ_{CDE}	95% CI
S1 Sequential dependence	24	8	17.5%	[6.7, 32.1]
S2 Necessary → sufficient	27	9	6.9%	[1.9, 12.6]
S3 Elimination over closed set	24	8	17.5%	[7.3, 28.5]
S4 Causal attribution	30	10	3.7%	[0.0, 10.2]
S5 Comparison (1 variable)	30	10	7.7%	[1.7, 15.0]
S6 Simultaneous constraints	27	9	14.6%	[6.3, 24.4]

Table 5: Δ_{CDE} by abstract inference schema. Descriptively, the range across schemas (3.7%–17.5%) is wider than across problem types (8.8%–13.4%); schema CIs overlap substantially. CIs from whole-triplet bootstrap (10,000 replicates).

Problem type	Items	Zero-load	Probed	Grounding
Computational	54	44	10	98.1% (53/54)
Theoretical	54	0	54	79.6% (43/54)
History/literature	54	0	54	70.4% (38/54)
Total	162	44	118	82.7% (134/162)

Table 6: Factual premise recall performance of the focal model across problem types.

J Schema-Level Corruption Dependence

Table 5 reports Δ_{CDE} stratified by abstract inference schema, with whole-triplet bootstrap confidence intervals (10,000 resamples, seed 20261013).

K Knowledge Probes and Factual Grounding

Knowledge probes evaluate standalone factual recall of external premises required to solve the benchmark questions but omitted from the problem stem. Table 6 summarizes focal-model factual competence across problem types.

44 of 54 computational items carry zero external knowledge load (100% grounding by construction); the remaining 10 are probed. Across items requiring external factual recall, the model demonstrates high grounding in physical formulas ($p = \rho gh$), stoichiometry (1 : 1 ratio for SO_2 and Br_2), biological transport, and historical

milestones (August 1945 general uprising). Substantive errors concentrate in granular humanities memorization (colonial decrees, modern poetry lines); 15 further probes received empty responses and are counted as failures. Failing items do not propagate more than passing ones (Δ_{CDE} 10.5% vs. 13.4%). Free-text probe answers were graded by an LLM judge (`deepseek/deepseek-chat`; 6 judge failures were overturned by expert audit). An independent second grading of all 258 focal-model probe answers found no wrong non-empty answer accepted by the judge; an item passes only if every one of its probes is judged correct.

References

[1] Alvan R Feinstein and Domenic V Cicchetti. 1990. High agreement but low kappa: I. The problems of two paradoxes. *Journal of Clinical Epidemiology* 43, 6 (1990), 543–549.